USER GUIDE

Valencia Stock Energy Workbench

How the tool works, which data it uses at every step, how to read every number it produces — and, just as importantly, what it cannot do.

01 · FILES
Four inputs
Validate and activate the GIS, Tipo15, climate and template the engine reads.

→

02 · RUN
One model per building
Pick a scope, read the preflight, then simulate a full year, building by building.

→

03 · OUTPUTS
Results and evidence
Totals, per-cluster comparison, a searchable ledger and the raw EnergyPlus files.

Example run in this guide
Benicalap district · `benicalap_v8`

Verified profile
`80b3d4dafcdb7190…`

Engine
EnergyPlus 25.2.0 · OpenStudio 3.11.0

Installation
See `getting-started.md`

1 What this is — and what it is not

The Workbench turns the Valencia cadastre into an energy result. It reads the official building dataset, constructs a separate OpenStudio model for *each individual building*, runs a full annual EnergyPlus simulation on every one of them, and reports the outcome per building, per typology cluster, per district and for the whole city.

It is a **local desktop application**. Nothing is uploaded; the simulation runs on your own machine and every file it produces stays there.

> **The honest one-line description**
> A normative stock estimate whose HVAC component is simulated — verified building-by-building against an independent author's own published EnergyPlus model, and *not* validated against measured consumption.

That sentence carries three claims, and each one is deliberate:

- **Normative.** Lighting, appliances and domestic hot water come from Spanish CTE standard densities, not from measurements. Together they are 90.1 % of the reported energy (§8).
- **Simulated HVAC.** Heating and cooling are genuinely computed by EnergyPlus from geometry, envelope, weather, shading and occupancy — that is the remaining 9.9 %.
- **Verified, not validated.** We reproduce a reference model to within 1.5 % (§9). Nobody has yet compared any of this to a utility meter.

Read §6 and §7 before using a number in a paper or a decision. They are short, and they are the part of this document that matters most.

CONTENTS

2 The three tabs at a glance

The whole application is three screens, used in order. Everything else that exists in the codebase is hidden from normal navigation (§10).

TAB	WHAT YOU DO	WHAT IT PRODUCES	WHEN YOU NEED IT
Files	Check that the four inputs are `ACTIVE`; upload replacements if you have newer data.	A frozen, hashed input set and a `RUN READY` state.	Once after installing, then only when data changes.
Run	Choose a scope, press *Run preflight*, read the counts, press *Start run*.	One EnergyPlus simulation per building, written to a durable ledger.	Every time you want new results.
Outputs	Select a finished run; read totals, compare clusters, search the ledger, open a building's files.	Aggregate figures, CSV table, signed evidence packages.	Whenever you are interpreting or reporting results.

You can start at Outputs

The distribution ships with a completed Benicalap run (`benicalap_v8`: 967 buildings, 4.4 GB of preserved EnergyPlus output). You can explore every screen in §5 without simulating anything.

3 Files — the four authoritative inputs

Everything the engine knows comes from four files. There are no hidden databases and no online lookups. Each card on the Files screen shows `ACTIVE` or `REQUIRED`, and the page header says `RUN READY` only when all four resolve.

What each file contributes

INPUT	FIELDS THE ENGINE ACTUALLY READS	WHAT IT DECIDES
Building GIS Valencia cadastral layer	`refparcela` · footprint geometry · `altura_max` · `cluster` · `pob_total` · `num_vivend` · district name	Which buildings exist, their shape and height, their typology, and how many people live in each.
Tipo15 companion Cadastral dwelling records	`31_pc` (parcel) · `252_planta` (floor) · `428_uso` (use) · `442_sup_Residencial` (area)	How much recorded dwelling area a parcel really has, and whether its ground floor is residential or commercial.
Climate pair EPW + DDY, activated together	8,760 hourly rows and site location · winter and summer design days	The weather the year is simulated against, and the extreme days the heat pumps are sized on.
PlantillaOS template OpenStudio library model	24 required roles: constructions, schedules and space types	Materials, occupancy and equipment schedules, thermostat setpoints — the physics library every model is assembled from.

> **Nothing is invented when data is missing**
> If a building has no Tipo15 record, the engine does not guess a dwelling area — it falls back to geometry and records which source it used in that building's row (`res_area_source`). If a file fails validation on upload, it is rejected rather than partially accepted.

Why the hashes matter

Each file is copied into managed storage and fingerprinted. A run resolves the active set *once, at the moment it starts*, and stores those fingerprints in every row it writes. Uploading a newer GIS file tomorrow therefore cannot silently rewrite what yesterday's results mean — and a run cannot be resumed against different inputs, because the identity would no longer match (§4).

4 Run — what happens to each building

This is the part that is usually invisible. For every single building in scope, the engine performs the following chain and then runs a complete 8,760-hour EnergyPlus simulation. The coloured squares show which of the four inputs each step consumes.

Building GIS | Tipo15 | Climate | Template

1	**Footprint** The cadastral polygon is simplified. If simplification would move its area too much, the building is excluded rather than distorted.	Building GIS
2	**Storey count, capped by cadastral evidence** Height comes from the GIS, but is limited to the storeys the recorded dwelling area can actually fill, plus one commercial ground floor.	Building GIS, Tipo15
3	**Envelope by typology and period** The building's TABULA cluster selects its walls, roof and floor — layer by layer, from real materials in the template.	Building GIS, Template
4	**Windows, frames and blinds** Discrete openings are placed per façade with balcony doors, aluminium frames, and Valencian *persianas* that close on high summer sun.	Template
5	**Neighbours: shading and party walls** Every real building within 50 m is loaded as shading mass; shared walls become adiabatic instead of losing heat to the street.	Building GIS
6	**Real occupancy from the population register** The registered resident count replaces the standard 20 m²/person norm — this changes both internal gains and hygienic ventilation.	Building GIS, Template
7	**Domestic hot water** A hot-water loop sized on the Spanish CTE figure of 28 litres per person per day — so it scales with the real residents, not with floor area.	Building GIS, Template
8	**Heat pumps, sized on design days** One packaged terminal heat pump per zone, autosized against the winter and summer design days from the DDY file.	Climate
9	**Annual simulation, then quality control** 8,760 hours of EnergyPlus. Afterwards the result is checked against the model's own geometry and the error file is scanned before the row is written.	Climate

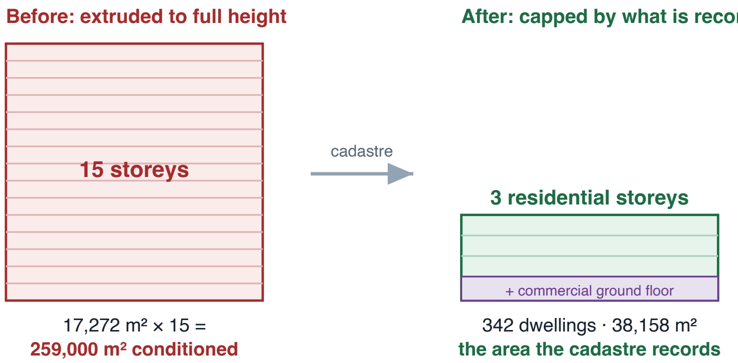
Each row is written to disk and flushed immediately — which is why a stopped run can always be resumed.

The per-building pipeline. Stage 2 is the one added most recently and it is explained below; it is the difference between the model conditioning 2.76 and 3.84 million m² across this district.

Stage 2 in detail: why storeys get capped

A model is built by extruding the footprint upward by the recorded height. On a parcel that covers a whole city block those two numbers describe *different things*: the footprint is the entire block, while the height is that of its tallest corner. Multiplying them invents floor area that does not exist.

EXAMPLE · PARCEL 3748901YJ2734H · BENICALAP



Across Benicalap

399 of 967 buildings were capped, carrying 55 % of the district's energy.

521 buildings came out bit-for-bit identical — they were already short, or had no cadastral record to check against.

342 dwellings × ~112 m² = 38,158 m² — the cadastre is internally consistent. Before the cap, this single building accounted for 5.6 % of the district's total energy.

> **What the cap does not fix**
>
> A capped building becomes shorter but keeps the full block footprint, so its shape is squat where the real building is a tower. Its surface-to-volume ratio is therefore wrong. Since heating and cooling are only 8.6 % of the total and lighting, appliances and hot water scale with area, the area correction dominates and the shape error is second-order — but it is real, and it is listed in §8.

Choosing a scope

SCOPE	USE IT FOR	BENICALAP EXAMPLE
Selected buildings	Testing, or reproducing one specific cadastral reference exactly.	seconds to minutes
One district	The normal working unit. Every eligible building in a named district.	967 buildings · 5.4 h
All Valencia	Production runs. Requires an extra explicit acknowledgement.	~24,900 buildings

Reading the preflight

Run preflight asks the engine itself — not an estimate — how many buildings it can actually process, and why it is dropping the rest. Exclusions are always given a reason:

REASON	BENICALAP	PLAIN MEANING
`simplification_area_change…`	39	The outline is so intricate that simplifying it would change the floor area beyond tolerance. Excluded rather than distorted.
`footprint_outside_range`	3	Smaller than 50 m² or larger than 20,000 m² — outside the range the single-zone-per-storey scheme is trusted in.
`duplicate_refparcela`	1	The same reference appears twice in the source data.
`interior_rings`	1	The polygon has a courtyard hole, which the extrusion cannot represent.

> **Count the loss by area, not by building**
> In this district 45 of 1,012 buildings produced no result — 4.4 % by count, but only **1.9 % by floor area**. Exclusions concentrate in unusual, often large-footprint buildings, so a headcount overstates the coverage gap in one direction and understates its importance in the other. Outputs always shows both.

Stopping and resuming

Every building's row is written and flushed to disk the moment it finishes, so the ledger is never behind the process. *Stop safely* lets the running buildings finish and then halts. *Resume same run* skips everything already completed and continues.

Resume refuses to continue if the inputs, the policy or the engine sources have changed since the run started. That is deliberate: half a run computed one way and half another is not a result, and the tool will not produce one.

5 Outputs — how to read every number

Select a finished run at the top of the screen. Six headline figures appear:

TOTAL SITE ENERGY **109.76** GWh/year · modelled subset	GEOMETRIC EUI **46.9** kWh/m² · modelled residential floor	CADASTRAL EUI **50.9** kWh/m² · Tipo15 dwelling area
CARBON **34,723** tCO_2/year · total site	COVERAGE **95.6 %** 967 OK · 45 excluded	QA STATUS **PASS** 0 QA failed · 0 unexplained severe

The single most misread number: which floor area?

Energy per square metre depends entirely on which square metres you divide by, and this building stock offers three defensible answers. The Workbench shows all three rather than picking one silently.

ONE BUILDING, THREE WAYS TO COUNT ITS FLOOR

Geometric
the residential floor the model built
46.9
kWh/m²

Cadastral
the dwelling area the cadastre records (Tipo15)
50.9
kWh/m² — used for "Δ vs Rai"

Conditioned
every zone the engine actually simulated
42.2
kWh/m² — the physical intensity

Why the first two divide by residential area only
The commercial ground floor is heated and lit, so its energy is in the numerator — but the reference study keeps it out of the denominator. We follow that convention so the comparison stays like-for-like.

Same building, same energy, three legitimate intensities. Always say which one you are quoting.
Benicalap totals: **2,340,150 m²** geometric · **2,157,843 m²** cadastral · **2,757,102 m²** conditioned.

Correction on record. Until 8 August 2026 the published note described the residential basis as "the floor the model actually conditions". That was false — the conditioned floor is a different and larger number — and the wording has been withdrawn.

The cluster table and "Δ vs Rai"

Buildings are grouped into the 21 TABULA typology clusters (type × construction period). For each, the table shows both denominators next to the published reference intensity and the difference. Benicalap contains 18 of the 21:

CLUSTER	BUILDINGS	CADASTRAL EUI	REFERENCE	Δ
`BlocPluriP04`	460	54.4	46.6	+16.9 %
`BlocPluriP05`	209	46.7	46.6	+0.2 %
`BlocPluriP03`	83	60.6	51.7	+17.2 %
`VivUniP02`	55	73.6	55.8	+31.9 %
`BlocPluriP06`	33	47.0	45.0	+4.4 %
`EdiPluriP02`	24	65.5	52.0	+26.0 %

What Δ vs Rai is, and what it is not

It is the distance between our per-building simulation and the constants published in the reference doctoral thesis, which were produced by simulating three representative geometries and assigning the result to every building in a cluster. It is agreement *with another model*. It is not agreement with measured consumption, and it cannot be used as evidence of accuracy.

The building ledger

Below the totals, every building is listed and searchable. Click a row to open its evidence drawer. Beyond the obvious columns, these are the fields that explain *why* a building came out the way it did:

FIELD	WHAT IT TELLS YOU
`storey_cap_applied`	Whether cadastral evidence actually reduced this building's geometry.
`conditioned_to_cadastral_ratio`	How much more floor the model conditions than the cadastre records. Near 1.0 is healthy; above 2 the result deserves scrutiny.
`occupancy_plausibility`	Flags implausible population density — for example 5.7 m² per person — without blocking the run.
`large_footprint_single_zone`	Marks buildings deep enough that one well-mixed zone per storey is a weak assumption.
`res_area_source`	Whether the dwelling area came from Tipo15 or fell back to geometry.
`severes_unexplained`	EnergyPlus severe messages that are *not* the one documented benign case. Anything here invalidates the row.
`qa_all_passed`	The post-simulation cross-check of area, glazing, zone count and unmet hours.

The files kept for every building

FILE	OPEN IT WHEN YOU WANT TO...
`eplustbl.htm`	see the full EnergyPlus report — end uses, surfaces, loads, sizing — in the same format as any other EnergyPlus study.
`qa_report.txt`	check what was verified after the simulation and by how much each check passed.
`model_python.osm`	open the actual model in OpenStudio and inspect or modify it yourself.
`eplusout.err`	read every warning EnergyPlus raised for this building.

Taking results out

- **Building CSV** — the whole ledger as a table, for a spreadsheet or a statistics package.
- **Signed building package** — one building: its model, preserved outputs and ledger row.
- **Full signed ZIP** — the entire run. It first measures the real file count and size and asks you to confirm, then writes a ZIP64 archive with an Ed25519-signed manifest so a recipient can prove nothing was altered in transit.

6 What the product can do

- CAN Simulate a full year for **each individual building** in the Valencia cadastre — not a typological average scaled up.
- CAN Use each building's **real registered population**, real footprint, real neighbours and real recorded dwelling area.

- CAN Run one building, one district or the whole city through the same engine, with no change in method.
- CAN Survive interruption: a stopped run resumes exactly where it left off, and refuses to resume under changed inputs.
- CAN Show the same result on three floor-area bases so a comparison is never accidentally unlike-for-unlike.
- CAN Preserve the complete EnergyPlus evidence for every building and export it signed.
- CAN Accept newer input data — a newer cadastre, a different climate file, a modified template — without any code change.
- CAN Prove its own integrity: it reconstructs an independent published building and scores itself against it (§9).

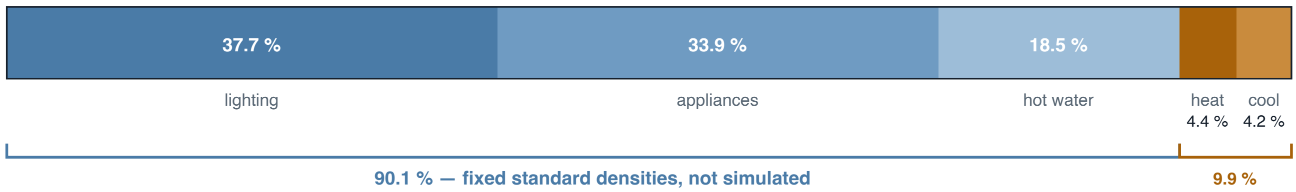
7 What it cannot do

- CANNOT **Tell you what a building actually consumes.** Nothing here has been calibrated against a meter. It reports what the standards and the physics imply, not what happened.
- CANNOT **Give an uncertainty band.** The sampling study exists but was run on an earlier configuration, so the current figures carry no confidence interval.
- CANNOT **Evaluate retrofit scenarios from these three tabs.** Scenario machinery exists in the codebase (§10) but is not part of this surface.
- CANNOT Split heating and cooling between the dwellings and the commercial ground floor. EnergyPlus meters them by end use, not by space type, so `residential_site_kwh` is an upper bound rather than a clean subtotal.
- CANNOT Represent a deep floor plate properly. Above the threshold each storey is one well-mixed zone with no core/perimeter distinction.
- CANNOT Model more than one weather year, or a warmed future climate — a single measured-year file is used throughout.
- CANNOT Justify its carbon figures. The fuel mix of the real stock is unconfirmed, so the tCO_2 number rests on an assumption nobody has verified.
- CANNOT Reconstruct the true shape of a capped block — it gets the area right and the massing approximately.

8 Known limitations, measured

Every limitation below has been quantified rather than described. The point of this table is that you can decide for yourself whether a given limitation matters for your question.

Where the energy actually comes from



This is the most important single fact about the result. Lighting, appliances and hot water follow Spanish standard densities and scale with floor area and occupants. Only heating, cooling, fans and pumps are the product of building physics. Work on insulation, shading or thermal bridges therefore moves under a tenth of the headline number — which does not make it worthless, but does tell you what the headline is sensitive to.

LIMITATION	MEASURED SIZE	WHY IT IS STILL OPEN
Lighting and appliances use a fixed standard density	71.6 % of site energy	No real appliance survey exists for this stock. The reference study uses the same assumption, so comparisons stay consistent.
Heating/cooling cannot be attributed by space type	8.6 %	EnergyPlus meters by end use. Splitting it needs an additional output table and a rerun.
Capped buildings keep a squat massing	399 buildings	The true vertical shape of a block-covering parcel is not recorded anywhere in the source data.
The smallest model is ground + 1, so short buildings cannot shrink further	74 buildings still >2× cadastre	Changing the minimum touches the frozen builder that all historical results depend on.
One well-mixed zone per storey above the footprint threshold	9 buildings · 8.0 % of energy	Flagged per building rather than silently applied, so the affected share is always visible.
Coverage of the district	96 % by count 98 % by area	The excluded buildings are genuinely unrepresentable, not skipped for convenience.
Buildings with zero registered residents get zero hot water	policy: `literal_zero`	A deliberate choice matching the reference thesis. The alternative policy exists and can be selected.
Fuel mix of the real stock is unconfirmed	all carbon figures	Waiting on information the research group has not yet supplied.

9 How to check we are not lying to you

A model that cannot be audited is an opinion. Four mechanisms make this one checkable, and all of them are available to you rather than to us alone.

It scores itself against an independent building

The tool reconstructs a building published by the reference author — from his own reported surface areas — and compares thirteen quantities against his numbers. Any failure makes the command exit non-zero.

QUANTITY	OURS	REFERENCE	DIFFERENCE
Residential floor area (m^2)	954.800	954.800	0.00 %
Wall U-value (W/m^2K)	1.409	1.409	0.00 %
HVAC sensible heating (GJ)	26.527	27.300	−2.85 %
HVAC sensible cooling (GJ)	−68.403	−65.030	+5.19 %
Total site energy (kWh/m^2)	**53.680**	**54.500**	**−1.50 %**

Read this honestly

The reconstruction shares the reference author's template and weather file, and is built from the surface areas he reported. It therefore verifies the *engine and the parameter set* — it does not independently confirm that the typological abstraction, or the wider stock geometry, is right.

The other three mechanisms

- **A profile fingerprint.** Every locked parameter and the checksum of every engine source file are hashed into one identifier (`80b3d4dafcdb7190…`) that is written into every result row.
- **A drift guard.** If any engine source changes, runs stop before starting and name the parameter that moved. A run therefore cannot span two versions of the physics.
- **Self-describing rows.** Each building's row carries the fingerprints of the inputs, the policy and the engine that produced it, so a single row can be traced without the rest of the run.

Reproducing an older run

When the engine is later corrected the fingerprint is retired. A result produced under a retired fingerprint stays valid and fully traceable — every row still names its own origin — but reproducing it byte-for-byte means checking out the source state that fingerprint points to, not the newest version.

10 Beyond the three tabs

The three-tab surface is deliberately small. Two other paths exist for anyone who wants them, and neither is needed for normal use.

- **Command line, one building.** `python src/verified_model.py --refparcela <REFERENCE>` runs the identical pipeline for a single building and writes the same evidence directory. `--verify` reruns the thirteen checks in §9; `--show-profile` prints every locked parameter.
- **Earlier research screens.** A geometry builder, a full OpenStudio editor, scenario controls, the representative-cluster neighbourhood and city pipelines and the sampling study are all still in the

application. They are hidden from navigation but reachable by direct URL. They belong to the earlier method and are kept for reproducing historical results, not for new work.

11 Glossary

TERM	MEANING
`refparcela`	The cadastral reference that uniquely identifies a parcel. The primary key everywhere in this tool.
Cluster	A TABULA typology class combining building type (detached, small block, large block) with construction period P01–P07. 21 exist in Valencia.
EUI	Energy Use Intensity — annual energy divided by floor area, in kWh/m². Meaningless unless the floor area basis is stated.
Terciario	The tertiary/commercial space type used for ground floors. It is *not* a terrace or roof level.
Tipo15	The cadastral companion file listing dwellings floor by floor: how many, what use, how much area.
`pob_total`	Registered residents per building, from the municipal population register (*padrón*) — actual residence, not design occupancy.
`ConsumE`	The reference study's published consumption intensity, one constant per cluster. The basis for the "Δ vs Rai" column.
EPW / DDY	The annual hourly weather file, and the extreme design days used to size equipment.
PTHP	Packaged Terminal Heat Pump — the heating and cooling unit modelled in each zone.
Preflight	A dry run in which the engine reports what it can and cannot process, before any simulation begins.
Ledger	The append-only file holding one row per building, flushed to disk immediately, which makes runs resumable and auditable.